

Cartesian products disprove asymptotic cube optimality

A negative answer to Problem 4.2 of arXiv:1311.4955

Autonomous solve-first investigation — lane 12

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Abstract

For a centrally symmetric convex body $K \subset \mathbb{R}^n$, put $P_n(K) = |\Pi K|/|K|^{n-1}$ and let S_n be the supremum of P_n . Saroglou asks whether $(S_n/2^n)^{1/n}$ tends to one. The answer is no. The functional P is exactly multiplicative under orthogonal Cartesian products, while the three-dimensional cross-polytope $O = B_1^3$ satisfies $P_3(O) = 9 > 8 = P_3(C_3)$. Hence $P_{3m}(O^m) = 9^m$, already giving an exponential obstruction. In fact, $S_n/2^n$ is supermultiplicative, so Fekete's lemma and the known exponential upper bound show that the full limit exists and is at least $(9/8)^{1/3} > 1$.

1 The source problem

For a convex body $K \subset \mathbb{R}^n$, its projection body is defined by

$$h_{\Pi K}(u) = \text{vol}_{n-1}(K \mid u^\perp) = \frac{1}{2} \int_{S^{n-1}} |\langle u, v \rangle| dS_K(v).$$

The affine-invariant functional in Schneider's projection problem is

$$P_n(K) = \frac{|\Pi K|}{|K|^{n-1}}.$$

For a cube C_n , $P_n(C_n) = 2^n$. After recalling fixed-dimensional counterexamples to cube maximality, Saroglou asks in Problem 4.2 whether the cube is nevertheless asymptotically optimal in exponential scale [1, p. 9]. The exact question is reproduced below.

Let us discuss another question concerning Schneider's problem. As mentioned earlier, there exists a convex body K with $P(K) > P(C_n)$. It is natural to ask however if Schneider's conjecture is in some sense "almost" correct. It is well known that

$$P(K) < A^n,$$

for all convex bodies K , where $A > 0$ is some absolute constant. What appears to be unknown is the following

Problem 4.2. *Is it true that the ratio*

$$\left(\frac{\max_K P(K)}{P(C_n)} \right)^{\frac{1}{n}}$$

tends to 1, as n tends to infinity? Here K runs over all symmetric convex bodies.

Write

$$S_n = \sup\{P_n(K) : K \subset \mathbb{R}^n \text{ is a centrally symmetric convex body}\}.$$

Thus the question is whether $(S_n/2^n)^{1/n} \rightarrow 1$.

2 Idea of the proof

The missing operation is Cartesian product. Cauchy's projection formula shows that the projection body of $K \times L$ is a correspondingly scaled Cartesian product of ΠK and ΠL . All volume factors cancel in P , making P multiplicative. Therefore a strict gap in any one fixed dimension repeats independently in every Cartesian factor and becomes an exponential gap.

The smallest convenient exact gap is three-dimensional. The projection body of the regular octahedron B_1^3 is a rhombic dodecahedral zonotope. Four determinants give its volume as 16, and hence $P_3(B_1^3) = 16/(4/3)^2 = 9$. Cartesian powers compare 9^m with the cube's 8^m .

3 Product multiplicativity

Lemma 1 (projection body of a product). *Let $K \subset \mathbb{R}^a$ and $L \subset \mathbb{R}^b$ be convex bodies, with the two spaces embedded orthogonally in $\mathbb{R}^{a+b}$. Then*

$$\Pi(K \times L) = |L| \Pi K \times |K| \Pi L \quad (1)$$

and consequently

$$P_{a+b}(K \times L) = P_a(K)P_b(L). \quad (2)$$

Proof. First suppose that K and L are polytopes. Every facet of $K \times L$ is either $F \times L$, where F is a facet of K , or $K \times G$, where G is a facet of L . Its outer normal is respectively $(u_F, 0)$ or $(0, u_G)$, and its area is respectively $|L| |F|$ or $|K| |G|$. Cauchy's formula therefore gives, for $(x, y) \in \mathbb{R}^a \oplus \mathbb{R}^b$,

$$h_{\Pi(K \times L)}(x, y) = |L| h_{\Pi K}(x) + |K| h_{\Pi L}(y).$$

The right side is the support function of the Cartesian product in (1). Arbitrary convex bodies follow by polytope approximation and continuity of projection bodies and volume.

Taking $(a+b)$ -dimensional volume in (1) gives

$$|\Pi(K \times L)| = |L|^a |K|^b |\Pi K| |\Pi L|.$$

After division by $|K \times L|^{a+b-1} = |K|^{a+b-1} |L|^{a+b-1}$, the powers of $|K|$ and $|L|$ cancel exactly, yielding (2). $\square$

4 The exact three-dimensional gap

Lemma 2. *For $O = B_1^3 = \text{conv}\{\pm e_1, \pm e_2, \pm e_3\}$,*

$$|\Pi O| = 16, \quad |O| = \frac{4}{3}, \quad P_3(O) = 9.$$

Proof. The eight facets of O are indexed by $s \in \{\pm 1\}^3$. Each has area $\sqrt{3}/2$ and outer unit normal $s/\sqrt{3}$. Hence Cauchy's formula gives

$$h_{\Pi O}(u) = \frac{1}{4} \sum_{s \in \{\pm 1\}^3} |\langle u, s \rangle| = \frac{1}{2} \sum_{j=1}^4 |\langle u, v_j \rangle|,$$

where one may take

$$v_1 = (1, 1, 1), \quad v_2 = (1, -1, -1), \quad v_3 = (-1, 1, -1), \quad v_4 = (-1, -1, 1).$$

Therefore

$$\Pi O = \sum_{j=1}^4 [-v_j/2, v_j/2].$$

For a three-dimensional symmetric zonotope $\sum_j [-g_j, g_j]$, its volume is $2^3 \sum_{i < j < k} |\det(g_i, g_j, g_k)|$; this follows by the standard dissection into the parallelotopes generated by triples. Every one of the four determinants $\det(v_i, v_j, v_k)$ has modulus 4. With $g_j = v_j/2$, each determinant therefore has modulus $4/2^3 = 1/2$, and

$$|\Pi O| = 2^3 \cdot 4 \cdot \frac{1}{2} = 16.$$

Finally, O is the union of eight coordinate tetrahedra of volume $1/6$, so $|O| = 4/3$ and

$$P_3(O) = \frac{16}{(4/3)^2} = 9.$$

$\square$

5 Negative answer and existence of the limit

Theorem 3. *Problem 4.2 has a negative answer. More precisely, the limit in the problem exists and*

$$\lim_{n \rightarrow \infty} \left(\frac{S_n}{2^n} \right)^{1/n} \geq \left(\frac{9}{8} \right)^{1/3} > 1. \quad (3)$$

Proof. Lemma 1 and central symmetry of Cartesian products imply $S_{a+b} \geq S_a S_b$. Since $P_n(C_n) = 2^n$, the sequence

$$T_n = \frac{S_n}{2^n}$$

is supermultiplicative and satisfies $T_n \geq 1$. Thus $b_n = \log T_n$ is superadditive. The source records an absolute constant A with $P_n(K) < A^n$ for every convex body, so b_n/n is bounded above. Fekete's lemma now yields

$$\lim_{n \rightarrow \infty} \frac{b_n}{n} = \sup_{n \geq 1} \frac{b_n}{n}, \quad \lim_{n \rightarrow \infty} T_n^{1/n} = \sup_{n \geq 1} T_n^{1/n}.$$

By Lemma 2, $S_3 \geq 9$, whereas $P_3(C_3) = 8$. Therefore $T_3 \geq 9/8$, which proves (3).

Equivalently, without invoking the existence upgrade, Lemma 1 already gives

$$P_{3m}(O^m) = 9^m, \quad \left(\frac{S_{3m}}{2^{3m}} \right)^{1/(3m)} \geq \left(\frac{9}{8} \right)^{1/3}.$$

Hence even the subsequence of dimensions divisible by three is uniformly separated from one. $\square$

6 Verification, novelty check, and scope

The companion script `code/verify_projection_product.py` checks all four determinants exactly over the integers, the zonotope volume 16, the octahedron volume $4/3$, the value $P_3(O) = 9$, and the finite Cartesian-power identities. The script is diagnostic; the proof above is fully exact and self-contained.

Cheap run-index searches and bounded exact-phrase/topic searches on 13 August 2026 found the source paper and later descriptions of the still open exact Schneider projection problem, but no cited resolution of Problem 4.2 or this Cartesian-product amplification. This is a bounded novelty check, not a priority claim.

Theorem 3 settles only the asymptotic Problem 4.2. It does not determine S_n , classify maximizers, resolve Schneider's exact projection problem, or settle the distinct polar projection-body Problem 4.5 in the source.

Human-review recommendation. Accept as a candidate full counterexample. The two substantive checks are the product identity (1) and the four determinant values in Lemma 2; both are elementary and exact.

References

- [1] C. Saroglou, *On the equivalence between two problems of asymmetry on convex bodies*, Discrete Comput. Geom. **54** (2015), 573–585; arXiv:1311.4955v2, Problem 4.2.